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Summary

AI / Forward Deployed Engineer with 5+ years shipping production LLM agent systems and the Snowflake/dbt data platforms that feed them. Builds multi-agent harnesses in LangChain/LangGraph and TypeScript, gated by hand-labeled eval sets, deterministic guardrails, and A/B testing in MLflow/DeepEval. Works directly with clients to turn ambiguous requirements into deployed systems

Work Experience

Becker's Healthcare | AI developer & Data Engineer

Jan 2025 - Present

- Built an LLM eval harness from 15 hand-labeled expert agent trajectories, scoring multi-turn runs with DeepEval and tracking results in MLflow; gated deployment on pass rate against the set, catching 11% of regressions before release.
- A/B-tested two tool-config variants of one agent against the fixed eval set, shipping the winner on a 17-point pass-rate improvement. Fine-tuned an open-weight model via LoRA on a domain-specific task, lifting eval accuracy from 55% to 83% over the un-tuned base.
- Implemented an evaluation framework with MLflow and DeepEval to assess agent performance via multi-turn regression and LLM-as-judge methodologies, establishing release gates and data-driven feedback loops.
- Drove integration of AI governance frameworks (ISO 42001, NIST AI RMF) into agent design, establishing automated guardrail enforcement and continuous compliance monitoring.
- Designed and built a production Retrieval-Augmented Generation (RAG) pipeline in n8n, orchestrating automated Apify web scraping, high-dimensional vector embeddings with Pinecone, and an LLM-driven style-validation scoring gate.
- Added a deterministic guardrail layer that validates every agent-proposed tool call against hard-coded business rules before execution, blocking 156 invalid actions over production runs.
- Engineered an 11-persona multi-agent orchestration framework (Clawd Bot, TypeScript) with a 4-tier intent-dispatch router, ~25 commands, an error/feedback learning loop, and a manual-approval human-in-the-loop gate.
- Took a vendor-built voice-automation stack (Retell AI + Twilio + CRM) from prototype to production-ready internal system, owning the integration across telephony and CRM endpoints and turning an ambiguous vendor handoff into a deployed, maintained service.
- Deployed and operated a multi-LLM agent framework (open-source hermes-agent) on AWS EC2 with systemd supervision, IP-locked security, and integration to read-only Snowflake and a knowledge graph.
- Architected a Snowflake-native ELT with a dbt medallion layer ingesting 25.5M+ PostHog events and 186K WordPress articles (dbt suite 25/25) on a scheduled Task DAG with auto-suspend and email alerting.
- Built custom automation triggers and code-level JavaScript processing pipelines within a hosted n8n platform to support automated lead capture funnels, multi-tier notification webhooks, and asynchronous follow-up workflows.

Decision Point Analytics | Program Manager

Jul 2022 - Dec 2022

- Integrated diverse data sources into warehouses and built Python ETL pipelines (cleaning, normalization, incremental loading) with SQL data modeling to streamline downstream workflows.
- Spearheaded the development of a Resource Management Tool in Power BI, improving transparency & project efficiency by 11% and implemented distributed data-processing workflows using Apache Spark with asynchronous task scheduling, reducing batch processing times by 50%.
- Optimized 60+ SQL queries using query optimization and indexing, applying normalization principles, reduction in execution time, developed database schemas using Microsoft Visio, applying entity-relationship modeling (ERD) and normalization techniques.
- Developed Power BI dashboards and self-service tools with DAX calculations and SQL integration, enabling senior management to track performance trends throughout the SDLC and supporting agile development cycles

VOLVO | Project Manager / Data Analyst

Jul 2019 - Jul 2022

- Managed project estimates and forecasts using Excel for modeling and scenario planning, and Python (Pandas, NumPy) for automated data validation, exceeding performance expectations by 5% and reducing data validation time by 17%, earning the 'Innovation Award'.
- Automated status of order reports for the production planning team using Excel VBA scripts and macros to merge, format, rename, separate, and sort columns, reducing daily report creation time by 30 minutes each morning.
- Utilized Power BI and SQL to build dashboards for 10+ manufacturing units, streamlining ETL processes and visualizing key metrics, which improved reporting turnaround time and helped stakeholders resolve project issues more quickly
- Crafted 20+ KPIs per assembly line using DAX, Power BI, and Python to track productivity targets, enabling informed decision-making on employee scheduling to ensure coverage of required productivity hours and enhance overall efficiency.

Indiana University | Teaching Assistant (Statistics)

Jul 2023 - Dec 2024

- Developed 20+ R and Python guides in Jupyter Notebooks automating data cleaning, transformation, and visualization, improving reproducibility for students and researchers.
- Utilized SQL for querying large datasets and Python (Pandas, NumPy) and R for implementing complex statistical models and data manipulation techniques, contributing to cancer research by enhancing the extraction and analysis of high-volume datasets.

Spirent Communication | Business Process Automation Intern

Jul 2023 - Dec 2024

- Implemented Approval Automation flow using Power Automate to streamline approval processes across various business units.
- Conducted exploratory data analysis (EDA) using Python (Pandas, NumPy), identifying bottlenecks and reducing cycle time by 25%.
- Developed Tableau dashboards to visualize KPIs and create heat maps, leading to a 20% reduction in approval processing time.

Projects

AI Vision + Deterministic Scoring Health App | [link](#)

- Built a health app pairing an AI vision model with a deterministic scoring layer, so subjective image analysis produces consistent, rule-based scores rather than free-form model output.
- Designed it on a scale-first data model built to absorb high write volume without schema rework as usage grows.

Live Multi-Model LLM Audit Engine | [link](#)

- Built a live audit engine that runs the same query across multiple LLMs and compares outputs to surface disagreement, hallucination, or drift between models.
- Implemented per-call failover so a failed or degraded model response transparently reroutes to a backup, keeping the audit available under provider outages.

Production RAG SaaS for Purchase-Order Extraction | [Link](#)

- Built a production RAG SaaS that extracts structured data from purchase orders across multiple LLMs, routing each document to the model best suited to its format.
- Engineered a self-training memory layer that captures corrections and feeds them back as retrieval context, so extraction accuracy improves on recurring vendor formats over time.

Statistical Modeling

- Built an R model (PCA, k-means, cross-validation) with a confusion-matrix evaluation framework, containerized in Docker and deployed to GCP AI Platform with Cloud Build CI/CD.

Social Media mining

- Implemented a RAG pipeline with LangChain, chunking Reddit content and using GPT-3.5 for context-aware summarization; prototyped an AgentCore + FastAPI orchestration layer to automate sentiment-driven topic recommendations.

Skills

- **Agent Engineering:** Agent Harnessing, AI Agents / Agent Building, Multi-agent orchestration, LangChain, LangGraph, AgentCore, RAG (Pinecone), Human-in-the-Loop, LLMs / LLM Application Development, Context Engineering, Generative AI, Multimodal AI Systems, Feature Engineering, API integrations
- **LLM Frameworks & Agent Platforms:** LangChain, LangGraph, Semantic Kernel, Mastra, AgentCore, Prompt Engineering / Prompt Iteration, OpenAI Agents SDK, OpenAI GPT, Claude, Gemini, Grok, Open Source Model
- **Evaluation & Governance:** A/B testing, LLM Observability Tools, CloudWatch, LLM-as-judge, LoRA fine-tuning, Langfuse, LangSmith, DeepEval, ISO 42001, NIST AI RMF, MLFlow, OpenTelemetry
- **Machine Learning & AI:** Predictive Modeling, Classification, Clustering, Data Mining, NLP, Model Fine-tuning
- **Programming Languages:** SQL, Python (Pandas, NumPy, Scikit-learn, Matplotlib), Matplotlib, R, TypeScript, Prompt Engineering / Prompt Iteration
- **Databases, Web Technologies & APIs:** MySQL, NoSQL, PostgreSQL, Neo4j, BigQuery, Snowflake, HTML5, JavaScript, FastAPI, OpenAPI, Swagger, AgentCore, AWS Neptune, Amazon API Gateway, Vector Databases
- **Cloud Platforms:** AWS, S3, EC2, Lambda, SageMaker, Google Cloud Platform, GCP, Microsoft Azure
- **DevOps & CI/CD:** Git, CI/CD Pipelines, GitHub Actions, Automated Deployments

Education

Indiana University <i>Master of Science, Information Science</i>	Jan 2023 - Dec 2024
S.G.S.I.T.S <i>Bachelor of Engineering, Electrical</i>	Apr 2015 - Jun 2019